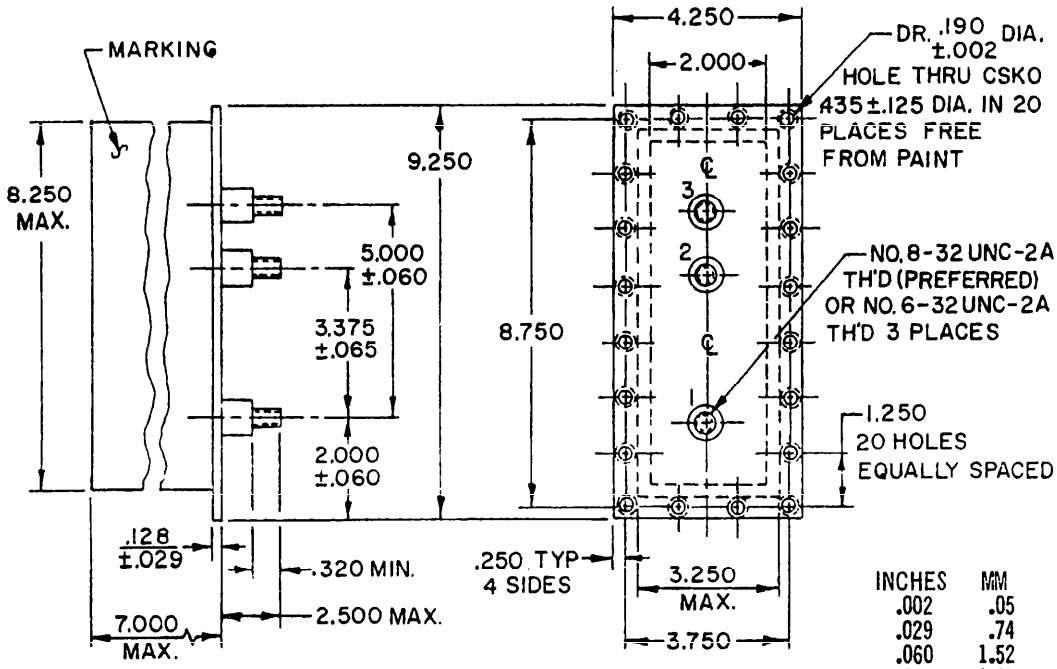
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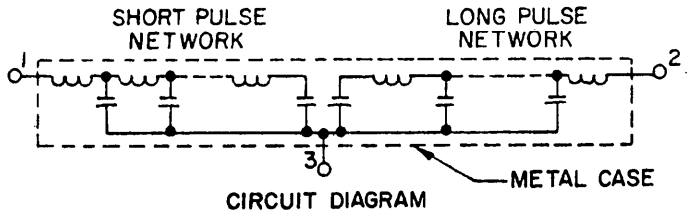
MILITARY SPECIFICATION SHEET

NETWORK, PULSE FORMING, RADAR, TYPE PN3E571J202Q500

The complete requirements for procuring the pulse forming network described herein shall consist of this document and the latest issue of Specification MIL-N-23182.



INCHES	MM
.002	.05
.029	.74
.060	1.52
.065	1.65
.125	3.18
.128	3.25
.190	4.88
.250	6.35
.320	8.13
.435	11.05
1.250	31.75
2.000	50.80
2.500	63.50
3.250	82.55
3.375	85.73
3.750	95.25
4.250	107.95
5.000	127.00
7.000	177.80
8.250	209.55
8.750	222.25
9.250	234.95



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).

Case and mounting dimensions, and circuit diagram.

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REQUIREMENTS:

Dimensions and configuration: See figure.

Case: Metal.

Circuit diagram: See figure.
Note - This item consists of two pulse forming networks in one container.

Weight: 14 pounds.

Maximum operating temperature: +85° C.

Pulse repetition frequency:
Long pulse network: 415 pulses per second.
Short pulse network: 2,000 pulses per second.

Total capacitance:
Long pulse network (between terminals 2 and 3): 0.025 uf ±5 percent.
Short pulse network (between terminals 1 and 3): 0.005 uf ±5 percent.

Load impedance: 50 ohms.

Supply voltage (to pulse forming networks): 8,000 vdc.

Charging voltage: 15,000 vdc.

Terminals: Screw.

Terminal strength: Method 211, test condition E, of MIL-STD-202. The following detail shall apply:
Torque: 11 pound-inches.

Sealing: As specified in MIL-N-23182. Networks shall be immersed for 2 to 3 minutes in a bath of water or any other liquid of no greater viscosity maintained at a temperature of at least +85° C. The temperature of the networks shall not exceed +40° C at the time of immersion. During the immersion, networks shall show no flow of air or gas bubbles (other than an occasional entrapped air or gas bubble), or leakage of compound.

Corona discharge: As specified in MIL-N-23182, at atmospheric pressures of from 29.92 inches of mercury (sea level) to 3.44 inches of mercury (50,000 feet) with no indications of corona.

Dielectric withstanding voltage: As specified in MIL-N-23182.
Pulse forming networks shall withstand the specified potentials without breakdown, puncturing of insulation, or arcing.

Barometric pressure (reduced): Method 105, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Application of test voltage: 125 percent of rated charging voltage for at least 1 minute between each terminal and every other terminal in turn, and between the container and each terminal not connected to the container.
After the test, networks shall be visually examined for evidence of short-circuiting, arcing, breakdown of insulation and other visible damage.

Insulation resistance: Method 302, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Electrification time: 1 minute, maximum.
Terminal to terminal (terminal 1 to terminal 3; and terminal 2 to terminal 3):
At 25° C: 500 megohm-microfarads, minimum.
At +85° C $^{+3^\circ}_{-0^\circ}$ C: 15 megohm-microfarads, minimum.
Terminals to container (where container is not a terminal), at 25° C: Shall exceed 3,000 megohms.

Dissipation factor at 25° C: As specified in MIL-N-23182. The following details and exceptions shall apply:
Frequency of measurement (or reference): 1,000 Hz 1/.
Test potential: 500 volts RMS, maximum.
Dissipation factor: Shall not exceed 1 percent.

Capacitance (terminal to terminal):
Method 305 of MIL-STD-202. The following details and exceptions shall apply:
Test frequency: 1,000 Hz., ±100 Hz.
Test potential: 500 volts RMS, maximum.
Limit of accuracy of measurement: ±2 percent.
Total measured capacitance:
Within the following limits:
Long pulse network:
0.02375 to 0.02625 uf.
Short pulse network:
0.00475 to 0.00525 uf.

Waveshape electrical characteristics:
Measured as specified in MIL-N-23182, (with exceptions noted below), and shall be as follows:

1/ Wherever used herein, the frequency measurement unit "Hertz (Hz)" is equivalent to and replaces "cycles per second (cps)".

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Long pulse network:
Pulse duration (at 70 percent amplitude): 2.65 μsec., ±0.15 μsec.
Percent ripple: 10 percent, maximum.
Pulse rise time: 0.28 μsec ±0.10 μsec.
Pulse decay time: 0.84 μsec ±0.10 μsec.
Characteristic impedance: 50 ohms.
Duty cycle: Between 0.0010375 and 0.001162.
Short pulse network:
Pulse duration (at 70 percent amplitude): 0.57 μsec ±0.05 μsec.
Pulse rise time: 0.12 μsec., ±0.05 μsec.
Pulse decay time: 0.36 μsec ±0.05 μsec.
Characteristic impedance: 50 ohms.
Duty cycle: Between 0.00104 and 0.00124.

Early life survival: As specified in MIL-N-23182. The following details and exceptions shall apply:
Prior to the test, networks shall be subjected to requirements, at standard test conditions, for corona discharge and capacitance.
Terminating resistance: Not to exceed 37.5 ohms.
Test temperature: Not less than +85° C.
Pulse repetition frequency:
Long pulse network: 104 pulses per second.
Short pulse network: 500 pulses per second.
At the conclusion of the 24-hour exposure, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge. In addition, capacitance shall be measured at 25° C. Capacitance change, from the initial measurement, shall not exceed ±2 percent.

Thermal shock: Method 107, test condition A, of MIL-STD-202. The following details and exceptions shall apply:
Exposure time:
At -55° $^{+0°}_{-3°}$ C: 2 hours.
At +85° $^{+3°}_{-0°}$ C: 2 hours.
After completion of the final cycle, networks shall be examined for evidence of leakage or other visible damage. In addition, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, capacitance, and waveshape electrical characteristics.

Moisture resistance: Method 106, figure 106-1, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: On racks by normal mounting means.
Conditioning: Networks shall undergo a 24-hour initial drying period prior to first cycle.
Polarization voltage: 100 vdc, during steps 1 to 6 inclusive between all terminals not connected directly to each other.
Upon completion of step 6 of the final cycle, networks shall be removed from the humidity chamber and shall be examined for evidence of leakage, or other visible damage. Networks shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insualtion resistance, capacitance, and waveshape electrical characteristics.

Vibration: Method 201 of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Planes of vibration: As specified in MIL-N-23182 for units weighing more than 10 pounds.
Measurements during vibration: During the last one-half hour of vibration, a suitable measurement shall be made to detect intermittent contacts, or open- or short-circuiting.
After the test, networks shall be examined for evidence of leakage, or mechanical damage. Networks shall then be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance, and capacitance.

Shock (medium impact): Method 205, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Method of mounting: By normal means.
Number and direction of applied shocks: 30 impact shocks, 10 in each of three mutually perpendicular planes.
After the test, networks shall be examined for evidence of physical or mechanical damage. In addition, networks shall be subjected to and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage barometric pressure (reduced), insulation resistance and capacitance.

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Low temperature: As specified in MIL-N-23182. Networks shall be subjected to a temperature of $-55^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$ C for 48 hours with rated charging voltage applied, with no evidence of breakdown.

Capacitance change at reduced temperature: As specified in MIL-N-23182. The following details and exceptions shall apply:
At the conclusion of the low temperature test, and while networks are still subjected to the -55° C temperature, the capacitance of the pulse forming networks shall be measured. The measurement shall be made at, or referred to, a frequency of 1,000 Hz. The test potential shall be 500 volts RMS or less. Capacitance change shall not exceed -5 percent.

Salt spray (corrosion): Method 101, test condition B, of MIL-STD-202. The following details and exceptions shall apply:
Salt solution concentration: 5 percent.
After the test, networks shall be examined for evidence of corrosion and obliteration of marking.

Life: As specified in MIL-N-23182 for networks with special life requirements. The following details and exceptions shall apply:
Exposure time: 1,000 hours.
Exposure temperature: $+85^{\circ}\ ^{+3^{\circ}}_{-0^{\circ}}$ C.
Test potential: 15,000 vdc.
Input power:
Long pulse network: 4,950 volt-amperes.
Short pulse network: 5,130 volt-amperes.
Temperature rise, as specified in MIL-N-23182, shall not exceed 30° C.
After the test, networks shall be subjected to, and shall comply with requirements, at standard test conditions, for corona discharge, dielectric withstanding voltage, barometric pressure (reduced), insulation resistance and waveshape electrical characteristics. In addition, networks shall be examined for evidence of physical or electrical damage.

Part number: Consists of the basic number of this specification sheet and a dash number.

M23182/5 -01

Basic document number

Dash number

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - MU, EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - MI, SM
Navy - MC, AS, CS
Air Force - None

Preparing activity:
Navy - SH

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